THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An element with configuration 2, 8, 7 forms an ion: 1

(a) X^{+} **(b)** X^{2-} **(c)** X^{-} **(d)** X^{2+}

2. Which of the following will decolourise bromine water? 1

(a) C_3H_8 **(b)** C_2H_6 **(c)** CH_4 **(d)** C_2H_4

3. Chips packets are filled with nitrogen to: 1

(a) increase the weight
(b) keep them warm
(c) make them react with air
(d) prevent oxidation of the oil in the chips

4. Which of the following is a strong base? 1

(a) $Al(OH)_3$ **(b)** NaOH **(c)** $Cu(OH)_2$ **(d)** $Mg(OH)_2$

5. The process of making oxide layer on aluminium thicker is: 1

(a) galvanisation **(b)** anodising **(c)** smelting **(d)** roasting

6. Carbon cannot form C_4^{-} because: 1

(a) 6 protons cannot hold 10 electrons easily
(b) it has 8 electrons
(c) it is metal
(d) it is heavy

7. The sum of coefficients in the balanced equation $Al + O_2 \longrightarrow Al_2O_3$ is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 9

8. The salt CH_3COONa in water is: 1

(a) amphoteric **(b)** acidic **(c)** basic **(d)** neutral

9. Which of the following is the correct order of reactivity? 1

(a) Fe > Cu > Mg > Zn **(b)** Zn > Mg > Cu > Fe
(c) Cu > Fe > Zn > Mg **(d)** Mg > Zn > Fe > Cu

10. The number of shared electron pairs in ethyne (C_2H_2) is: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

11. Which of these is both a combination and an exothermic reaction? 1

(a) $2\,H_2O \longrightarrow 2\,H_2 + O_2$ **(b)** $CaO + H_2O \longrightarrow Ca(OH)_2$ **(c)** $CaCO_3 \longrightarrow CaO + CO_2$ **(d)** $2\,AgCl \longrightarrow 2\,Ag + Cl_2$

12. Hardening of plaster of Paris is due to: 1

(a) decomposition **(b)** oxidation
(c) loss of water **(d)** formation of gypsum

13. The formula of the compound formed between Mg and Cl is: 1

(a) MgCl **(b)** $MgCl_2$ **(c)** Mg_2Cl **(d)** Mg_2Cl_2

14. The reaction $C_2H_4 + H_2 \longrightarrow C_2H_6$ with nickel is: 1

(a) combustion **(b)** substitution reaction
(c) addition reaction **(d)** esterification

15. Galvanised iron does not rust because zinc: 1

(a) combines with iron
(b) is lighter than iron
(c) is less reactive than iron
(d) is more reactive than iron and forms a protective layer

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are ionic compounds usually soluble in water but insoluble in kerosene? 2

17. Write the reaction of methanoic acid with sodium hydroxide. 2

18. Why is the combustion of methane an exothermic reaction? Write the balanced equation. 2

19. A solution has pH 2 and another has pH 4. Compare their H^+ concentrations. 2

20. Why is it that copper does not react with dilute HCl but zinc does? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Metal A reacts with dilute HCl but not B. A reacts with cold water but not C. Arrange A, B, C in order of reactivity and justify. 3

22. Explain why diamond and graphite have different properties though both are carbon. 3

23. Zinc granules are added to blue copper sulphate solution. Give the observation, the equation and the reason. 3

24. State what happens when plaster of Paris is mixed with water and explain why a cast hardens. 3

OR

Why is it important to refine metals? Describe how impure copper is refined. 3

25. Explain the formation of micelles in soap solution. 3

OR

Lead nitrate is heated in a boiling tube. State three observations. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. For 5

(i) $CuO + H_2 \longrightarrow Cu + H_2O$ and
(ii) $Fe_2O_3 + 3\,CO \longrightarrow 2\,Fe + 3\,CO_2$, identify the substance oxidised, reduced, and the oxidising and reducing agents.

OR

Describe the properties and formulae of acids and bases using equations: reaction with metals, carbonates and each other. 5

27. Explain the method of extraction of 5
(a) mercury
(b) zinc and
(c) sodium with reasons for the method.

OR

Compare ethane, ethene and ethyne in formula, bonds, combustion and reactions. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND